SAT-Style Practice Test – December Original Form

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SECTION 1, MODULE 1: Reading and Writing

Question 1
Skill: Words in Context

The following text is adapted from a memoir by Japanese American dancer Aiko Tanemura. Tanemura is describing her return to a village she had left as a child.

I arrived in Kurose just after sunset. My uncle was waiting near the ferry dock, but he stared past me at first, trying to ______ the grown woman in the traveler before him. I had been gone so long that even my voice seemed to surprise him.

Which choice completes the text with the most logical and precise word or phrase?

A) locate

B) provoke

C) conceal

D) imitate

Answer: A

Question 2
Skill: Words in Context (Phrase)

The ______ Camila Rios's novel *The Salt Orchard* has been unusually enthusiastic. Reviewers across several countries have praised its layered narration, and the book recently received a major international fiction prize.

Which choice completes the text with the most logical and precise word or phrase?

A) opposition to

B) reception of

C) disposal of

D) separation from

Answer: B

Question 3
Skill: Words in Context

Some civic theorists argue that a democracy cannot function well unless voters have a basic knowledge of the past. In their view, history is too ______ to public decision-making to be left only to professional historians.

Which choice completes the text with the most logical and precise word or phrase?

A) ornamental

B) critical

C) accidental

D) interchangeable

Answer: B

Question 4
Skill: Words in Context

At a conference on children's literature, a speaker noted that while many illustrators have created excellent work, Leila Morgan's illustrations for *The Lantern Fox* ______ all other examples of the form: according to the speaker, no picture book better combines movement, color, and humor.

Which choice completes the text with the most logical and precise word or phrase?

A) preceded

B) eclipsed

C) borrowed

D) postponed

Answer: B

Question 5
Skill: Words in Context

Jules Bastien-Lepage's painting *Haymaking* portrays exhausted farmworkers in plain clothing and muted light. The work ______ the polished conventions of many academic paintings of the period, which often presented rural labor as graceful, cheerful, and picturesque.

Which choice completes the text with the most logical and precise word or phrase?

A) rejected

B) preserved

C) measured

D) scheduled

Answer: A

Question 6
Skill: Function of an Underlined Phrase

In the early years of commercial aviation, airline executives expected passengers to embrace overnight flights quickly. But many travelers hesitated to buy tickets, **particularly after several newspapers published accounts of engine failures during storms**. Airlines therefore invested heavily in safety demonstrations and pilot-training publicity before expanding their night routes.

Which choice best describes the function of the underlined phrase in the text as a whole?

A) It identifies a reason some travelers were reluctant to use a new service.

B) It explains why airlines stopped developing overnight routes permanently.

C) It compares the safety of different forms of transportation.

D) It introduces evidence that travelers preferred newspaper advertisements.

Answer: A

Question 7
Skill: Function of an Underlined Portion

The following text is adapted from an 1892 short story. In the story, Mara Ilyin is a wealthy patron of young composers.

Any student from the conservatory who arrived at Mara Ilyin's apartment with a score folded beneath one arm could be sure of a hearing. **The timid and the confident, the brilliant and the merely ambitious, all found her door open**. She bought concert tickets, introduced promising musicians to conductors, and, when necessary, gave stern criticism.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It suggests that Mara Ilyin valued only confident musicians.

B) It creates a contrast between public concerts and private lessons.

C) It emphasizes the breadth of Mara Ilyin's support for inexperienced artists.

D) It explains why Mara Ilyin disliked most new musical compositions.

Answer: C

Question 8
Skill: Function of an Underlined Phrase

Argyroxiphium kauense is a plant in a group of Hawaiian species **known collectively as the silverblade alliance**. Members of this group vary widely in appearance: some are low shrubs, while others grow in tall rosettes on volcanic slopes. Biologists have traced these species to a single ancestral plant that reached the islands millions of years ago and then diversified into many habitats.

Which choice best describes the function of the underlined phrase in the text as a whole?

A) It lists all species in a plant group found on the Hawaiian Islands.

B) It supplies the name used to refer to a group of related plant species.

C) It explains why *Argyroxiphium kauense* grows only on volcanic slopes.

D) It provides the common name of the ancestral plant from which the group evolved.

Answer: B

Question 9
Skill: Main Idea

Textile artist Marisol Vega uses thread, fabric scraps, and hand-dyed wool to tell stories of resilience. Some of her pieces celebrate famous public figures, such as scientist Ellen Ochoa, while others portray ordinary workers, neighbors, and children. Vega has said that these approaches serve the same goal: showing how courage and connection can be found in both widely known lives and everyday ones.

Which choice best states the main idea of the text?

A) Vega's artwork recognizes both famous and ordinary examples of human strength and connection.

B) Vega began by portraying scientists but now focuses only on everyday people.

C) Vega's portraits of children are more technically advanced than her portraits of famous people.

D) Scholars have paid more attention to Vega's famous subjects than to her ordinary subjects.

Answer: A

Question 10
Skill: Main Idea

Community astronomy involves professional astronomers collaborating with members of the public. This approach can deepen public understanding by giving participants a direct role in scientific work. It can also greatly expand the amount of data available to researchers, as when Dr. Laila Chen and colleagues used thousands of sky-brightness readings submitted by students and amateur astronomers to study light pollution across several cities.

Which choice best states the main idea of the text?

A) Chen and colleagues were surprised that community astronomy changed participants' opinions about cities.

B) Community astronomy can help the public understand science while also allowing researchers to collect more data.

C) Community astronomy benefits researchers only when participants have advanced scientific training.

D) Light pollution studies are the only projects for which community astronomy is useful.

Answer: B

Question 11
Skill: Data Interpretation (Table)

[Table: "Names and Movements of Beetles during Sound Trials." Row 1: Beetle name Atlas; species *Scarabaeus viridis*; common name green scarab; direction of movement away from sound. Row 2: Beetle name Mina; species *Tenebrio aurum*; common name gold mealworm beetle; direction of movement toward sound. Row 3: Beetle name Nox; species *Carabus niger*; common name black ground beetle; direction of movement away from sound.]

Biologists exposed several beetles to a brief sound and recorded whether each beetle moved toward or away from it. Based on the table, a student concludes that Atlas and Nox behaved similarly in the sound trials.

Which choice best describes data from the table that support the student's conclusion?

A) Atlas moved toward the sound, while Nox moved away from it.

B) Mina and Atlas both moved away from the sound.

C) Atlas and Nox both moved away from the sound.

D) Nox moved toward the sound, while Mina moved away from it.

Answer: C

Question 12
Skill: Strengthen Claim

The Museo del Puerto in Valparaiso, Chile, is known for its maritime art collection. In 2021 the museum began offering free virtual tours. Some administrators worried that people who viewed the virtual tours would decide that visiting in person was unnecessary. However, after reviewing visitor surveys, the administrators argued that the virtual tours actually encouraged people to come to the museum.

Which statement, if true, would most directly support the administrators' claim?

A) Most surveyed visitors reported living in Valparaiso rather than in another city.

B) Many surveyed visitors said the virtual tours convinced them to plan an in-person visit.

C) Many surveyed visitors said they watched the virtual tours only after visiting in person.

D) Most surveyed visitors said they had never heard of the virtual tours before arriving.

Answer: B

Question 13
Skill: Textual Evidence (Quotation)

George Eliot's novel *Silas Marner* portrays the title character as a lonely weaver whose life changes when he begins caring for a child. The novel suggests that human affection can restore a person's connection to the community, as is evident when ______

Which quotation from *Silas Marner* most effectively illustrates the claim?

A) "The little child had come to link him once more with the whole world."

B) "The room was filled with the whirring of the loom."

C) "The gold had been carried away in the darkness."

D) "The village street was quiet under the winter sky."

Answer: A

Question 14
Skill: Logical Completion

A regional production of the musical *River Lanterns* opened in 2018 and was later revised for a national tour in 2022. In a review of the tour, critic Maya Greene praised the show's storytelling. Greene also explained that she had seen the earlier version in 2018 and had found its plot difficult to follow. This suggests that, in Greene's view, ______

Which choice most logically completes the text?

A) *River Lanterns* improved substantially between its 2018 production and its 2022 tour.

B) the 2018 version of *River Lanterns* had fewer storytelling problems than the 2022 version did.

C) *River Lanterns* should not have been revised before the national tour.

D) the national tour of *River Lanterns* was less enjoyable than the regional production.

Answer: A

Question 15
Skill: Logical Completion

Food scientists Dr. Priya Anand and Dr. Leon Fisk studied the frying of plantain chips. During frying, water in the outer layer evaporates, leaving tiny spaces that fill with oil and make the surface crisp. As frying continues, moisture from the center moves outward as long as the outer layer remains permeable. Therefore, the less moisture a chip loses during frying, ______ Which choice most logically completes the text?

A) the lower the oil's temperature must have been throughout the process.

B) the less crisp the chip's surface is likely to become.

C) the more plantain slices must have been placed in the fryer.

D) the faster the chip's outer layer must lose permeability.

Answer: B

Question 16
Skill: Verb Agreement

Water boils at about 212°F at sea level, but in Leadville, Colorado, where the elevation is over 10,000 feet, it boils at around 195°F. Food writer Lena Ortiz, who studies high-altitude cooking, ______ that lower boiling points can change the texture of pasta, rice, and vegetables.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) explain

B) explains

C) have explained

D) are explaining

Answer: B

Question 17
Skill: Verb Form

The White River Chapter is one of the 110 chapters of the fictional Red Mesa Nation. The chapter, known in the Diné language as *Tó Łichíí*, was the subject of a newspaper profile that ______ in the *Mesa Times* on June 14, 2019.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) appearing

B) has appeared

C) appeared

D) appears

Answer: C

Question 18
Skill: Punctuation and Possessive Form

With the documentary *Song of the Cedar*, Haida and Tlingit director Naomi Pierce joined a growing number of Indigenous women adding their voices to North American ______ short film, an official selection of the Pacific Northwest Film Festival, is about a carver and teacher in Sitka, Alaska.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) cinema, her

B) cinema. Her

C) cinema and her

D) cinema her

Answer: B

Question 19
Skill: Sentence Structure

Working with instruments designed for distances smaller than a human hair, ______

Which choice completes the text so that it conforms to the conventions of Standard English?

A) engineers have created sensors that can detect trace chemicals in water.

B) sensors that detect trace chemicals in water were created by engineers.

C) trace chemicals in water are detected by sensors created by engineers.

D) creating sensors that can detect trace chemicals in water.

Answer: A

Question 20
Skill: Punctuation (Appositive Phrase)

Just as a mathematical theorem can become associated not with its first discoverer but with the person who made it famous, the Cavendish balance in physics takes its name from a ______ who refined and publicized the device.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) natural philosopher: in this case, Henry Cavendish,

B) natural philosopher – in this case, Henry Cavendish –

C) natural philosopher in this case Henry Cavendish,

D) natural philosopher – in this case Henry Cavendish

Answer: B

Question 21
Skill: Punctuation (Conjunctive Adverb)

Breakwaters–long structures built offshore–are often constructed to reduce the force of incoming waves. Breakwaters can sometimes have the opposite ______ interrupting natural sand movement along a coast can increase erosion in nearby areas.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) effect, however,

B) effect, however

C) effect; however,

D) effect however

Answer: A

Question 22
Skill: Punctuation (Appositive)

The editors of *Field Notes: A Guide to Desert Plants* turned to historian Arturo Salazar to craft the entry for *ocotillo* but chose ______ Maya Lien to write the entry for *mesquite*.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) poet,

B) poet

C) poet:

D) poet;

Answer: B

Question 23
Skill: Transitions

Guard cells are specialized cells that form the tiny openings in a plant's leaves. These cells help regulate the amount of carbon dioxide a plant takes in. ______ they also help control how much water vapor the plant releases.

Which choice completes the text with the most logical transition?

A) In conclusion,

B) Additionally,

C) Instead,

D) Previously,

Answer: B

Question 24
Skill: Transitions

Himalia, one of the many known moons of Jupiter, is designated Jupiter VI. Roman numerals in moon designations usually indicate the order in which the moons were named rather than their distance from the planet. It would be incorrect to assume, ______ that Himalia is the sixth-closest moon to Jupiter.

Which choice completes the text with the most logical transition?

A) however,

B) then,

C) moreover,

D) finally,

Answer: B

Question 25
Skill: Rhetorical Synthesis (Emphasize Similarity)

While researching a topic, a student has taken the following notes:

- Dinosaur fossil specimens can be found in science museums around the world.
- A *Tyrannosaurus rex* fossil specimen nicknamed Rhea is housed at a museum in Alberta, Canada.

- Rhea lived during the Late Cretaceous period.
- A *Triceratops* fossil specimen nicknamed Vale is housed at a museum in Queensland, Australia.
- Vale lived during the Late Cretaceous period.

The student wants to emphasize a similarity between the two dinosaur fossil specimens. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both the *Tyrannosaurus rex* specimen Rhea and the *Triceratops* specimen Vale lived during the Late Cretaceous period.

B) Rhea is housed in Alberta, Canada, whereas Vale is housed in Queensland, Australia.

C) Dinosaur fossil specimens can be found in science museums around the world, including in Canada and Australia.

D) The Alberta museum houses Rhea, a *Tyrannosaurus rex* specimen from the Late Cretaceous period.

Answer: A

Question 26
Skill: Rhetorical Synthesis (Emphasize Similarity)

While researching a topic, a student has taken the following notes:

- Artist Grimanesa Amorós is known for large LED light sculptures.
- Her sculpture *Blue Tide* is made of smooth multicolored LED domes.
- *Blue Tide* occupies 300 cubic feet of space.
- Her sculpture *Current Web* is made of tangled blue and white LED tubes.
- *Current Web* occupies 90,000 cubic feet of space.

The student wants to emphasize a similarity between *Blue Tide* and *Current Web*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) At 90,000 cubic feet, *Current Web* occupies much more space than *Blue Tide*.

B) *Blue Tide* and *Current Web* are both LED light sculptures by Grimanesa Amorós.

C) The smooth LED domes of *Blue Tide* contrast with the tangled tubes of *Current Web*.

D) Grimanesa Amorós created *Blue Tide*, a sculpture made of smooth multicolored LED domes.

Answer: B

Question 27
Skill: Rhetorical Synthesis (Overview of Findings)

While researching a topic, a student has taken the following notes:

- In a 2022 study, researchers examined the rate at which 18 languages conveyed information.
- They measured information rate in bits per second.
- Information rate was approximately consistent across the 18 languages.
- The average was 40 bits per second.
- They also measured syllable rate, or syllables spoken per second.
- Italian had the third-fastest syllable rate at 7.8 syllables per second.
- Thai had the fifteenth-fastest syllable rate at 5.1 syllables per second.

The student wants to present an overview of the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Although some languages differed in syllable rate, all 18 conveyed information at roughly the same rate.

B) The 2022 study measured the information rate of 18 languages in bits per second.

C) Italian had a faster syllable rate than Thai, but the notes do not provide either language’s exact information rate.

D) Researchers found that information was conveyed more quickly in Italian, at 7.8 syllables per second, than in Thai, at 5.1 syllables per second.

Answer: A

SECTION 1, MODULE 2: Reading and Writing

Question 1
Skill: Words in Context

When governments design conservation policies for species at risk, lessons from earlier ecological crises can be ______. For instance, the factors that contributed to the extinction of several island birds in the nineteenth century may help scientists protect endangered island birds today.

Which choice completes the text with the most logical and precise word or phrase?

A) irrelevant

B) applicable

C) decorative

D) unstable

Answer: B

Question 2
Skill: Words in Context (Phrase)

The Atlantic sturgeon is an ectothermic fish, whereas the opah has partial endothermy, which allows portions of its body to stay warmer than the surrounding water. Researchers found that the salmon shark's body temperature remains 1.0 to 1.7°C above ambient water temperature, which is ______ classifying the salmon shark as a strict ectotherm.

Which choice completes the text with the most logical and precise word or phrase?

A) inconsistent with

B) subordinate to

C) convenient for

D) equivalent to

Answer: A

Question 3
Skill: Words in Context

In the late 2010s, prices for rare mechanical keyboards rose sharply, creating perfect conditions for ______ demand: buyers who had never previously cared about collecting keyboards entered the market, hoping prices would continue to rise and the keyboards could later be resold at a profit.

Which choice completes the text with the most logical and precise word or phrase?

A) suppressing

B) exploiting

C) eliminating

D) disguising

Answer: B

Question 4
Skill: Overall Structure

Spanning the 1930s to the 1990s, architect Lina Bardi's career moved through several distinct phases. As shown by the glass-and-concrete Casa de Vidro, many of her early projects embraced modernist simplicity. Later travels and collaborations, however, deepened her interest in vernacular building practices, and a more playful mixture of modern forms and local materials can be seen in projects such as the SESC Pompeia cultural center.

Which choice best describes the overall structure of the text?

A) It presents a broad claim about an architect's career, describes a design approach from an early phase, and then explains a later shift in that career.

B) It names a famous architect, rejects the main influence on her work, and then gives examples proving the rejection.

C) It provides two examples of an architect's projects and argues that the later project is less important than the earlier one.

D) It summarizes the architect's career and then claims that all phases of the career shared identical design principles.

Answer: A

Question 5
Skill: Function of an Underlined Statement

Individual mountain goats and Arctic herbivores such as caribou often follow predictable seasonal routes through their lifetimes. **This tendency has led some researchers to speculate that the extinct steppe bison may have done the same.** Yet chemical analysis of one preserved steppe bison horn shows that the animal's range expanded when it reached maturity and then contracted later in life.

Which choice best describes the function of the underlined statement in the text as a whole?

A) It introduces an inference based on a trait shared by certain animals, an inference that is later complicated by new evidence.

B) It proves that steppe bison and caribou used identical migration routes.

C) It explains why chemical analysis cannot reveal movement patterns in extinct animals.

D) It describes a behavior that the text later shows is absent from all living herbivores.

Answer: A

Question 6
Skill: Cross-Text Comparison

Text 1
French painter Edgar Degas often chose frames for his paintings that differed from the ornate gold frames common in official exhibitions. Like many artists of his circle, Degas believed that a frame should support a painting's colors and composition rather than distract from them.

Text 2
Painters associated with Impressionism frequently considered the color of a frame part of a painting's presentation. Some chose white or pale frames because those frames could make the colors of a painted scene appear more vivid, whereas gold frames could draw attention away from subtle shifts of light.

Based on the texts, both authors would most likely agree with which statement?

A) Many painters associated with Impressionism intentionally selected frames to affect how their works were viewed.

B) Degas's frames were always more expensive than the frames used in official exhibitions.

C) Gold frames were preferred by most Impressionist painters because they made colors brighter.

D) Frame color was less important to Impressionist painters than to artists in official exhibitions.

Answer: A

Question 7
Skill: Inference (Literature)

The following text is adapted from a translated novel. The narrator introduces the story of a young clerk named Tomas.

The story of Tomas, which we are about to tell, is not offered because Tomas was famous; indeed, almost no one outside his village knew his name. But the story itself deserves attention. Though it concerns a person of no great rank, it took place in a moment when ordinary choices carried unusual weight, and for that reason the account should be preserved.

What does the text most strongly suggest about the story of Tomas?

A) It is difficult to understand because Tomas was known only in his village.

B) It matters despite Tomas's lack of fame because the events themselves are significant.

C) It should be ignored because stories about ordinary people cannot be historically useful.

D) It became important only after Tomas achieved national recognition.

Answer: B

Question 8
Skill: Data Interpretation (Table)

[Table: "Monthly Temperatures and Wing Centroid Sizes of Moth Specimens." Columns: Month; average high (°F); average low (°F); average male wing centroid size (mm); average female wing centroid size (mm). Row 1: April, 68, 45, 1.82, 2.06. Row 2: June, 84, 61, 1.89, 2.17. Row 3: August, 91, 69, 1.93, 2.21. Row 4: October, 70, 47, 1.83, 2.08.]

In a study of moths, researchers measured wing centroid size as a proxy for adult life span: larger centroid size is associated with a longer life span. The researchers suggest that moths collected in warmer months tend to have longer life spans.

Which choice most effectively uses data from the table to support the researchers' suggestion?

A) The average male wing centroid size was larger in August than in April.

B) The average female wing centroid size was always larger than the average male wing centroid size.

C) The average monthly low temperature was higher in October than in April.

D) The average female wing centroid size was 1.93 mm in August and 2.08 mm in October.

Answer: A

Question 9
Skill: Data Interpretation (Graph)

[Graph: "Maximum Positive Electrical Charge for Three Pollinators." The y-axis is picocoulombs (pC), from 0 to 900. European moths have a maximum charge of about 20 pC; orchard honeybees have about 110 pC; red mason bees have about 760 pC.]

Heliandra plants typically carry a slight negative electrical charge, while pollinators tend to carry positive charges. A research team hypothesized that the difference in charges could attract the plant's stamens to pollinators if the pollinator's positive charge exceeds a certain threshold. The team found that orchard honeybees exceeded the threshold, which suggests that ______

Which choice most effectively uses data from the graph to complete the text?

A) red mason bees can also attract the stamens.

B) European moths and red mason bees both cannot attract the stamens.

C) European moths tend to repel the stamens.

D) the threshold positive charge must be greater than 760 pC.

Answer: A

Question 10
Skill: Strengthen Conclusion

Growing seasons in parts of Alaska have lengthened in response to climate change. Longer seasons may increase carbon dioxide absorption by marsh plants, but they may also increase carbon dioxide output through microbial respiration in soil. Hydrologist Nora Kim and colleagues modeled the effects of earlier spring snowmelt and concluded that net carbon dioxide uptake is likely to increase if warming causes snow to melt earlier.

Which finding, if true, would most directly support the researchers' conclusion?

A) Earlier snowmelt extends the time during which marsh plants can absorb carbon dioxide and has little effect on microbial respiration.

B) Earlier snowmelt slows plant growth by reducing soil insulation and strongly increases microbial respiration.

C) Earlier snowmelt reduces soil moisture needed by marsh plants and raises the rate of microbial respiration.

D) Earlier snowmelt increases both plant growth and microbial respiration by exactly the same amount.

Answer: A

Question 11
Skill: Strengthen Claim

A public transportation team in Detroit is considering a new streetcar stop in a neighborhood that currently lacks one. The team surveyed residents on sunny days and found that many said they would walk ten minutes to a streetcar stop. But a researcher argued that the survey may not accurately reflect residents' willingness to walk to a stop in everyday conditions.

Which finding, if true, would most directly support the researcher's claim?

A) The neighborhood has added many sidewalks and crosswalks in the last decade.

B) Residents are far less willing to walk to public transit on rainy or icy days than on sunny days.

C) The number of cyclists in the neighborhood has increased sharply.

D) Current streetcar riders are satisfied with the number of stops along the existing line.

Answer: B

Question 12
Skill: Strengthen Hypothesis

Some food packaging releases silver nanoparticles (Ag-NPs) that can enter waterways. A 2016 study found that Ag-NPs accumulate in the bodies of a marine worm, *Arenicola marina*. Because monitoring fish such as Atlantic salmon is costly, scientists hypothesize that measuring Ag-NP levels in *A. marina* could serve as a useful proxy for monitoring Ag-NP contamination in fish.

Which finding, if true, would most directly support the hypothesis?

A) When *A. marina* and Atlantic salmon are exposed to similar Ag-NP levels, individuals of the two species tend to accumulate similar amounts after adjusting for body size.

B) Ag-NP concentrations in *A. marina* vary widely even when environmental levels are stable.

C) *A. marina* can tolerate Ag-NP concentrations that cause harmful effects in Atlantic salmon.

D) It is easier to detect harmless concentrations of Ag-NPs in *A. marina* than harmful concentrations in salmon.

Answer: A

Question 13
Skill: Logical Completion

To reduce deformation in materials used in high-performance engines, researchers tested several ceramic fibers. Compared with two standard fibers, a boron-treated silicon carbide fiber had the lowest creep rate, a measure of how quickly a material deforms under constant stress and high temperature. The finding suggests that ______

Which choice most logically completes the text?

A) unlike the standard fibers, the boron-treated fiber cannot be used when temperatures and loads are constant.

B) engine components containing the boron-treated fiber may withstand mechanical stress longer than components containing either standard fiber.

C) the two standard fibers have chemical properties that make them more resistant to deformation than the boron-treated fiber.

D) the boron-treated fiber likely increases the creep rate of materials used in high-performance engines.

Answer: B

Question 14
Skill: Logical Completion

Ecologist Chelsea Wood and colleagues studied temperature-driven changes in the abundance of parasites. One parasite that requires three host species decreased as sea temperatures rose, whereas several directly transmitted parasites remained stable. Wood's team noted that parasites with multiple hosts might be buffered against some environmental changes, but the observed decline suggests that ______

Which choice most logically completes the text?

A) the use of multiple host species does not necessarily protect a parasite from all effects of rising temperatures.

B) directly transmitted parasites are always more vulnerable to warming than parasites with three hosts.

C) three-host parasites cannot be affected by changes in any of their host species.

D) all parasites in the study declined at the same rate as temperatures increased.

Answer: A

Question 15
Skill: Logical Completion

Scientists created a model to predict how shorter winters will affect future mammal populations in US national forests. Unfortunately, when the model is applied to large forests, its predictions for large mammals are too high, and when it is applied to small forests, its predictions for small mammals are too high. Pine Ridge Forest is a small forest. If used to evaluate shorter winters there, the model would likely ______

Which choice most logically completes the text?

A) underestimate the effect of shorter winters on large mammals.

B) reflect only factors unrelated to shorter winters.

C) overestimate the population sizes of small mammals.

D) ignore all predator-prey relationships in the forest.

Answer: C

Question 16
Skill: Sentence Structure

The hypothesized ancestor from which all modern Kartvelian languages descended ______

Which choice completes the text so that it conforms to the conventions of Standard English?

A) the language's descendants' properties were used by experts to infer Proto-Kartvelian.

B) experts reconstructed Proto-Kartvelian by inferring its properties from those of its descendants.

C) Proto-Kartvelian has been reconstructed by experts, who inferred the language's properties from those of its descendants.

D) the experts who reconstructed Proto-Kartvelian inferred the properties from its descendants.

Answer: C

Question 17
Skill: Punctuation (Independent Clauses)

Gqom, a style of electronic dance music that originated in Durban, South Africa, has become an international phenomenon. In 2023, producer DJ Mvelo performed at major festivals in Europe and Asia, ______ becoming one of the first South African gqom artists to headline those events.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) although

B) and

C) thus

D) with

Answer: C

Question 18
Skill: Punctuation (Dash in Parenthetical Information)

The Austronesian language family includes more than 1,200 languages–including Tagalog and Malagasy, which are spoken by tens of millions of people–and accounts for one-fifth of the world's languages, ______ of great interest to linguists.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) making it

B) which makes them

C) made it

D) it is making

Answer: A

Question 19
Skill: Punctuation (Colon and Contrast)

Although 2060 Chiron and 2008 FC76 are both classified as centaurs, outer solar system bodies in unstable orbits, they exhibit striking differences in activity: ______ 2060 Chiron is considered an active centaur because it shows sporadic comet-like activity, 2008 FC76 shows no such activity and is considered dormant.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) for example,

B) while

C) therefore,

D) unless

Answer: B

Question 20
Skill: Verb Form

Mitochondrial genomes reproduce asexually, which should over time cause harmful mutations to accumulate. However, nuclear genes are hypothesized to evolve in ways that ______ the rate of mitochondrial decline.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had counteracted

B) counteracted

C) counteract

D) counteracting

Answer: C

Question 21
Skill: Subject-Verb Agreement

In a national election, one might expect campaigns to focus on the largest states. However, if polling data suggest that a smaller state could decide the final result, a campaign may invest heavily there. Ultimately, a state's voting record and polling data, not its population size, ______ its importance to campaign strategists.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) determines

B) has determined

C) determining

D) determine

Answer: D

Question 22
Skill: Transitions

To notice subtle variations in the progress of still-life painting, first consider Willem Kalf's *Still Life with a Silver Ewer* from 1655. ______ compare it to Rachel Ruysch's *Flowers on a Stone Ledge* from 1716.

Which choice completes the text with the most logical transition?

A) Therefore,

B) Still,

C) Instead,

D) Next,

Answer: D

Question 23
Skill: Transitions

Soil contaminated with lead is harmful to many plants and animals, but the plant species *Brassica juncea* not only survives such conditions but also absorbs lead into its roots and shoots. ______ lead concentrations in the soil decrease over time.

Which choice completes the text with the most logical transition?

A) Specifically,

B) Accordingly,

C) Nevertheless,

D) In addition,

Answer: B

Question 24
Skill: Rhetorical Synthesis (Provide an Example)

While researching a topic, a student has taken the following notes:

- The English word "dollar" comes from the German word "Thaler."
- Today, more than twenty different national currencies are referred to as "dollars."
- Singapore's currency is known as the Singapore dollar.

The student wants to provide an example of a country that uses the word "dollar" in the name of its currency. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Singapore uses the Singapore dollar as its national currency.

B) Singapore does not refer to its currency by the German word “Thaler.”

C) The German word “Thaler” is the origin of the English word “dollar.”

D) Today, many countries use the word “dollar” in naming their currencies.

Answer: A

Question 25
Skill: Rhetorical Synthesis (Emphasize Difference)

While researching a topic, a student has taken the following notes:

- Most freshwater mussels in Lake Marna are native species.
- In a 2020 study, researchers examined the role of birds in moving mussel larvae.
- *Lampsilis aurora*, a native mussel, was found attached to fish gills in four samples.
- *Dreissena pallida*, a nonnative mussel, was found attached to boat hulls in nine samples.
- The researchers concluded that boats likely contribute to the spread of nonnative mussels.

The student wants to emphasize a difference between the two mussel species. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both *Lampsilis aurora* and *Dreissena pallida* were included in a 2020 study of mussels.

B) Although *Lampsilis aurora* is native to Lake Marna, *Dreissena pallida* is not.

C) Researchers studied freshwater mussels in Lake Marna and examined how they move.

D) Boats likely contribute to the spread of nonnative mussels in Lake Marna.

Answer: B

Question 26
Skill: Rhetorical Synthesis (Emphasize Order)

While researching a topic, a student has taken the following notes:

- A supercontinent is a single landmass containing most or all of Earth’s continents.
- Supercontinents form and later break apart over hundreds of millions of years.
- Columbia was a supercontinent that formed about 1.8 billion years ago.
- Pangaea was a supercontinent that formed about 335 million years ago.

The student wants to emphasize the order in which the supercontinents formed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Columbia formed about 1.8 billion years ago, long before Pangaea formed about 335 million years ago.

B) Columbia and Pangaea were both supercontinents, single landmasses containing most or all continents.

C) Supercontinents form and break apart over hundreds of millions of years.

D) Pangaea and Columbia each eventually broke apart after forming.

Answer: A

Question 27
Skill: Rhetorical Synthesis (Audience-Aware Summary)

While researching a topic, a student has taken the following notes:

- In a 2004 study, researchers examined how fallen leaves affect seedling emergence.
- Seedling emergence is when a seedling sprouts aboveground and begins photosynthesis.
- The study tested a dry grassland in the central United States.
- In those conditions, fallen leaves had a positive effect on seedling emergence.
- The study was conducted by Hart and Kim.

The student wants to present the study's finding to an audience already familiar with seedling emergence. Which choice most effectively uses information from the notes to accomplish this goal?

A) Seedling emergence occurs when a seedling sprouts aboveground and begins photosynthesis.

B) Hart and Kim found that in a dry central US grassland, fallen leaves had a positive effect on seedling emergence.

C) Fallen leaves are plant matter that may collect on the ground in grassland environments.

D) The study by Hart and Kim was published in 2004.

Answer: B

SECTION 2, MODULE 1: Math

Question 1
Skill: Linear Equations in Context

A triangle has one side with length *a* centimeters and two sides each with length *b* centimeters. The perimeter of the triangle is 68 centimeters. Which equation represents this situation?

A) $a + b = 68$

B) $a + 2b = 68$

C) $2a + b = 68$

D) $2a + 2b = 68$

Answer: B

Solution: The perimeter is the sum of the three side lengths: $a + b + b = a + 2b$*, so* $a + 2b = 68$*.*

Question 2
Skill: Linear Equations

If $7 + x = 4$, what is the value of $20 + 5x$?

Answer: 5

Solution: $7 + x = 4$*, so* $x = -3$*. Then* $20 + 5x = 20 + 5(-3) = 5$*.*

Question 3
Skill: Rates and Proportions

A recipe for snack mix uses 4 cups of oats and 7 handfuls of dried fruit and contains a total of 1,840 milligrams (mg) of potassium. There are 320 mg of potassium in 1 cup of oats. How much potassium, in mg, is in 1 handful of dried fruit?

A) 560

B) 160

C) 80

D) 40

Answer: C

Solution: The oats provide 4(320) = 1,280 mg. The dried fruit provides 1,840 - 1,280 = 560 mg total. Each of the 7 handfuls has 560/7 = 80 mg.

Question 4
Skill: Percentages

A bookstore is offering all notebooks at a 25% discount off their original price. What is the discount, in dollars, on a notebook with an original price of $64?

A) 16

B) 25

C) 48

D) 80

Answer: A

Solution: 25% of 64 is 0.25(64) = 16.

Question 5
Skill: Evaluating Expressions

What is the value of 4(3x - 5) when x = 6?

A) 13

B) 28

C) 52

D) 72

Answer: C

Solution: 4(3(6) - 5) = 4(18 - 5) = 4(13) = 52.

Question 6
Skill: Systems of Linear Equations

3x = 24
-4x + y = -15

The solution to the given system of equations is (x, y). What is the value of x + y?

A) -17

B) -1

C) 9

D) 25

Answer: D

Solution: From 3x = 24, x = 8. Then -4(8) + y = -15, so -32 + y = -15 and y = 17. Thus x + y = 8 + 17 = 25.

Question 7
Skill: Linear Inequalities in Context

A customer is buying pens that cost $1.20 each. The customer has a coupon for $4 off the entire purchase and wants to spend no more than $20 before tax. Which inequality represents the number of pens, x, that the customer can purchase?

A) 1.20x + 4 <= 20

B) 1.20x - 4 <= 20

C) 1.20x - 4 >= 20

D) 4x - 1.20 <= 20

Answer: B

Solution: The cost after the coupon is 1.20x - 4. Since the customer wants to spend no more than $20, 1.20x - 4 <= 20.

Question 8
Skill: Area of a Circle

Circle M has a radius of 18 centimeters. What is the area of circle M, in square centimeters?

A) 18pi

B) 36pi

C) 162pi

D) 324pi

Answer: D

Solution: The area of a circle is pi r^2. With r = 18, the area is pi(18)^2 = 324pi.

Question 9
Skill: Circle Equations

In the xy-plane, circle A is defined by (x - 4)^2 + (y + 7)^2 = 36. Circle B has the same center as circle A, and the radius of circle B is 5 units greater than the radius of circle A. Circle B is defined by (x - 4)^2 + (y + 7)^2 = p, where p is a constant. What is the value of p?

Answer: 121

Solution: Circle A has radius sqrt(36) = 6. Circle B has radius 6 + 5 = 11, so p = 11^2 = 121.

Question 10
Skill: Combining Like Terms

Which expression is equivalent to 7mn^2 + 4mn + 5mn^2?

A) 12mn^2 + 4mn

B) 12mn^4 + 4mn

C) $16m^{2n}3$

D) 35mn^4

Answer: A

Solution: Combine like terms: 7mn^2 + 5mn^2 = 12mn^2. The expression is 12mn^2 + 4mn.

Question 11
Skill: Equivalent Ratios

If x/y = 96 and cx/(6y) = 96, what is the value of c?

Answer: 6

Solution: cx/(6y) = (c/6)(x/y) = (c/6)(96). Since this equals 96, c/6 = 1, so c = 6.

Question 12
Skill: Similar Triangles

In the figure shown, triangle ABC is similar to triangle ADE. The measure of angle ADE is 58°, and BC = 12(DE). What is the measure of angle ABC?

A) 12°

B) 46°

C) 58°

D) 696°

Answer: C

Solution: Corresponding angles of similar triangles are congruent. Angle ABC corresponds to angle ADE, so angle ABC = 58°.

Question 13
Skill: Interpreting Points on a Graph

The graph shows the estimated boiling point y, in degrees Celsius, of a solution with salt concentration x percent, where 0 <= x <= 25. Which statement is the best interpretation of the point (12.5, 104.1)?

A) A solution with a boiling point of 12.5°C has a salt concentration of 104.1%.

B) A solution with a salt concentration of 12.5% has an estimated boiling point of 104.1°C.

C) The minimum estimated boiling point is 104.1°C.

D) The maximum salt concentration is 12.5%.

Answer: B

Solution: In an ordered pair (x, y), x represents salt concentration and y represents boiling point. Thus (12.5, 104.1) means 12.5% concentration and 104.1°C.

Question 14
Skill: Evaluating Functions

The function g is defined by g(x) = 3x^2 - 10x + 1. What is the value of g(4)?

A) -1

B) 5

C) 9

D) 21

Answer: C

Solution: g(4) = 3(4^2) - 10(4) + 1 = 48 - 40 + 1 = 9.

Question 15
Skill: Rearranging Equations

If 96/x = 8x, what is the value of 8x^2?

A) 768

B) 96

C) 12

D) 8

Answer: B

Solution: Multiply both sides by x: 96 = 8x^2. Therefore, 8x^2 = 96.

Question 16
Skill: Linear Inequalities

Which point (x, y) is a solution to the inequality y <= 4x - 9 in the xy-plane?

A) (-2, 4)

B) (0, 9)

C) (3, 2)

D) (4, 12)

Answer: C

Solution: For (3, 2), 4(3) - 9 = 3, and 2 <= 3 is true. The other choices do not satisfy the inequality.

Question 17
Skill: Systems of Inequalities in Context

A florist is arranging rose stems and lily stems. The florist will use x rose stems and y lily stems. The arrangement must include at least 14 stems total and cost no more than $46. Each rose stem costs $2, and each lily stem costs $5. Which system of inequalities represents this situation?

A) x + y >= 14 and 2x + 5y <= 46

B) x + y <= 14 and 2x + 5y <= 46

C) x + y >= 14 and 5x + 2y <= 46

D) x + y <= 14 and 5x + 2y >= 46

Answer: A

Solution: At least 14 stems gives x + y >= 14. The cost is 2x + 5y and must be no more than 46, so 2x + 5y <= 46.

Question 18
Skill: Solving for a Variable

The equation relates the positive numbers f, x, and g:

f = (6x)^2/(7g)

Which equation correctly expresses x in terms of f and g?

A) x = sqrt(7fg)/6

B) x = 6sqrt(7fg)

C) x = sqrt(7fg/6)

D) x = sqrt(7fg) - 6

Answer: A

Solution: f = 36x^2/(7g), so 7fg = 36x^2. Thus x^2 = 7fg/36, and because x is positive, x = sqrt(7fg)/6.

Question 19
Skill: Parallel Lines

In the xy-plane, line n passes through (0, 0) and is parallel to the line represented by $y = 9x - 2$. If line n also passes through (7, d), what is the value of d?

Answer: 63

Solution: Parallel lines have the same slope. Line n has slope 9 and passes through the origin, so its equation is $y = 9x$. At $x = 7$, $y = 63$, so $d = 63$.

Question 20
Skill: Circle Equations

A circle in the xy-plane has center (11, -6) and radius 4k. Which equation represents this circle?

A) $(x - 11)^2 + (y + 6)^2 = 16k^2$

B) $(x + 11)^2 + (y - 6)^2 = 16k^2$

C) $(x - 11)^2 + (y + 6)^2 = 4k^2$

D) $(x - 11)^2 + (y - 6)^2 = 4k$

Answer: A

Solution: A circle with center (h, k) and radius r has equation $(x - h)^2 + (y - k)^2 = r^2$. Here $r = 4k$, so $r^2 = 16k^2$, and the center is (11, -6), giving $(x - 11)^2 + (y + 6)^2 = 16k^2$.

Question 21
Skill: Trigonometry in Context

A ramp must form an angle theta with level ground such that $\tan(\theta) \le 1/10$. If a ramp conforms to this rule and has a height of 36.7 inches, what is the least possible horizontal distance x, in inches?

Answer: 367

Solution: $\tan(\theta)$ = height/horizontal distance = $36.7/x$. The least x occurs when $36.7/x = 1/10$. Thus $x = 36.7(10) = 367$.

Question 22
Skill: Exponential Functions

The equation of a graph is $y = a^x + b$, where a and b are constants. The graph passes through (0, 2) and (1, 8). What is the value of $a - b$?

Answer: 6

Solution: At (0, 2), $2 = a^0 + b = 1 + b$, so $b = 1$. At (1, 8), $8 = a + 1$, so $a = 7$. Therefore, $a - b = 7 - 1 = 6$.

SECTION 2, MODULE 2: Math

Question 1
Skill: Evaluating Functions

f(x) = (x + 12)/4

For the function f defined by the equation shown, what is the value of f(-8)?

A) -5

B) -1

C) 1

D) 5

Answer: C

Solution: f(-8) = (-8 + 12)/4 = 4/4 = 1.

Question 2
Skill: Mean

32, 33, 34, 35, 36, 39, 43

What is the mean of the data set shown?

A) 33

B) 35

C) 36

D) 41

Answer: C

Solution: The sum is 32 + 33 + 34 + 35 + 36 + 39 + 43 = 252. The mean is 252/7 = 36.

Question 3
Skill: Exponential Decay / Graph Interpretation

The voltage across a capacitor is initially 20 volts and decreases over time, as shown in a graph. Which of the following is closest to the time, in seconds, from the initial measurement at which the voltage is 10% of its initial value?

[Graph: an exponential decay curve that begins at 20 volts when time is 0 and reaches 2 volts at approximately 2.30 seconds.]

A) 0.69

B) 1.15

C) 2.30

D) 4.60

Answer: C

Solution: 10% of 20 volts is 2 volts. The graph indicates that the voltage reaches 2 volts at about 2.30 seconds.

Question 4
Skill: Lines of Best Fit

A scatterplot shows the relationship between x and y for data set A. A line of best fit is shown with a y-intercept of approximately 61.25. Data set B is created by subtracting 7 units from the y-value of each data point in data set A. Which value is closest to the y-coordinate of the y-intercept of the line of best fit for data set B?

A) 54.25

B) 61.25

C) 68.25

D) 75.25

Answer: A

Solution: Subtracting 7 from every y-value shifts the line down by 7, so the new y-intercept is 61.25 - 7 = 54.25.

Question 5
Skill: Linear Functions

In the linear function f, f(0) = -3 and f(9) = -3. Which equation defines f?

A) f(x) = -3

B) f(x) = 0

C) f(x) = x - 3

D) f(x) = 9x - 3

Answer: A

Solution: Since f has the same value at x = 0 and x = 9 and is linear, its slope is 0. Therefore f(x) = -3.

Question 6
Skill: Linear Functions from Tables

For the linear function f, the table shows values of x and f(x).

x	f(x)
2	-9
6	-29
11	-54

Which equation defines f?

A) f(x) = -5(x - 1)

B) f(x) = -5(x + 1)

C) f(x) = -4(x + 1)

D) f(x) = -5x + 1

Answer: D

Solution: The slope is (-29 - (-9))/(6 - 2) = -20/4 = -5. Using (2, -9), -9 = -5(2) + b, so b = 1. Thus f(x) = -5x + 1.

Question 7
Skill: Function Interpretation

A sample of a substance takes 18 years to decay to half its original mass. The function s(t) = 96(0.5)^(t/18) gives the approximate mass, in grams, remaining t years after a 96-gram sample starts to decay. Which statement is the best interpretation of s(36) = 24 in this context?

A) Approximately 24 grams of the sample remains 36 years after the sample starts to decay.

B) The mass has decreased by 36 grams 24 years after the sample starts to decay.

C) Approximately 36 grams of the sample remains 24 years after the sample starts to decay.

D) The sample takes 24 years to decay to 36 grams.

Answer: A

Solution: s(36) = 24 means that when t = 36 years, the remaining mass is 24 grams.

Question 8
Skill: Conditional Probability

The table shows the distribution of people in a town by age group.

Age group	Proportion
Less than 18 years old	28%
18-39 years old	31%
40-64 years old	25%
Greater than 64 years old	16%

If a person in this town is selected at random, which of the following is closest to the probability that the person is greater than 64 years old, given that the person is at least 18 years old?

A) 0.16

B) 0.22

C) 0.41

D) 0.72

Answer: B

Solution: P(at least 18) = 31% + 25% + 16% = 72%. P(greater than 64 and at least 18) = 16%. The conditional probability is 16/72 = 0.222..., closest to 0.22.

Question 9
Skill: Volume of a Cone

A right circular cone has a base diameter of 18 inches and a height of 10 inches. The volume of the cone is kpi cubic inches. What is the value of k?

Answer: 270

Solution: The radius is 9 inches. The volume is (1/3)pi r^2h = (1/3)pi(9^2)(10) = (1/3)pi(810) = 270pi. Thus k = 270.

Question 10
Skill: Triangle Midsegment Theorem

In triangle ABC, point D is the midpoint of AB and point E is the midpoint of BC. If BD = 9 and DE = 16, what is the length of AC?

A) 18

B) 25

C) 32

D) 50

Answer: C

Solution: Since D and E are midpoints, DE is a midsegment parallel to AC, and DE = AC/2. Thus AC = 2(16) = 32.

Question 11
Skill: Linear Equations and Number of Solutions

x(r - 9) + 5 = 14x + 31

In the given equation, r is a positive integer. If the equation has exactly one solution, what CANNOT be the value of r?

A) 5

B) 9

C) 20

D) 23

Answer: D

Solution: x(r - 9) + 5 = 14x + 31 gives x(r - 9 - 14) = 26, or x(r - 23) = 26. If r = 23, the equation becomes 0 = 26, which has no solution. For exactly one solution, r cannot be 23.

Question 12
Skill: Percentages in a Distribution

In a collection of tickets, 15% are red, 20% are green, 40% are blue, and 25% are yellow. If there are 36 green tickets, how many blue tickets are in the collection?

A) 45

B) 54

C) 72

D) 90

Answer: C

Solution: If 20% of the collection is 36, the total is 36/0.20 = 180. Blue tickets are 40% of 180, which is 72.

Question 13
Skill: Quadratics and the Discriminant

x(11x + 19) + c = 6

In the given equation, c is a constant. A solution to the equation is (-19 + sqrt(229))/22. What is the value of c?

A) 9

B) 12

C) 17

D) 21

Answer: A

Solution: The equation is 11x^2 + 19x + c - 6 = 0. By the quadratic formula, x = (-19 ± sqrt(19^2 - 4(11)(c - 6)))/22. Since the discriminant is 229, 361 - 44(c - 6) = 229. Thus 44(c - 6) = 132, c - 6 = 3, and c = 9.

Question 14
Skill: Polynomial Factors

One of the factors of 4x^3 + 56x^2 + 160x is x + b, where b is a positive constant. What is the smallest possible value of b?

Answer: 4

Solution: Factor the polynomial: 4x^3 + 56x^2 + 160x = 4x(x^2 + 14x + 40) = 4x(x + 4)(x + 10). The smallest positive b is 4.

Question 15
Skill: Exponential Growth in Context

A model estimates that at the end of each year from 2018 to 2023, the number of foxes in a preserve was 140% more than the number at the end of the previous year. The model estimates that at the end of 2019, there were 192 foxes. Which equation represents this model, where n is the estimated number of foxes t years after the end of 2018 and t <= 5?

A) n = 80(1.4)^t

B) n = 80(2.4)^t

C) n = 192(1.4)^t

D) n = 192(2.4)^t

Answer: B

Solution: 140% more means multiply by 2.4 each year. If t = 1 at the end of 2019, then 192 = n0(2.4), so n0 = 80. Therefore n = 80(2.4)^t.

Question 16
Skill: Quadratic Equations in Geometry

A rectangular garden is surrounded by a stone path that is x feet wide on all sides. The garden is 18 ft long and 10 ft wide. The area of the stone path is 288 ft^2. What is the value of x?

A) 2

B) 4

C) 11

D) 22

Answer: B

Solution: The outer rectangle has dimensions 18 + 2x and 10 + 2x. The garden area is 180. The path area is (18 + 2x)(10 + 2x) - 180 = 288. Thus (18 + 2x)(10 + 2x) = 468. Expanding gives 180 + 56x + 4x^2 = 468, so 4x^2 + 56x - 288 = 0. Divide by 4: x^2 + 14x - 72 = 0 = (x + 18)(x - 4). Since x is positive, x = 4.

Question 17
Skill: Linear Relationships with Parameters

The table shows values of x and y, where s is a constant. There is a linear relationship between x and y.

x	y
-3s	21
-s	15
2s	6

Which equation represents this relationship?

A) $3x + sy = 12s$

B) $sx + 3y = 12$

C) $3x + sy = 12$

D) $sx + 3y = 12s$

Answer: A

Solution: The slope from (-3s, 21) to (-s, 15) is (15 - 21)/(-s - (-3s)) = -6/(2s) = -3/s. Using point (2s, 6): y - 6 = (-3/s)(x - 2s), so y = (-3/s)x + 6 + 6 = (-3/s)x + 12. Multiplying by s gives sy = -3x + 12s, or 3x + sy = 12s.

Question 18
Skill: Inequalities with Consecutive Integers

In a set of four consecutive odd integers ordered from least to greatest, the first integer is x. The product of 18 and the third odd integer is at most 42 less than the sum of the first and fourth odd integers. What is the greatest possible value of x?

Answer: -7

Solution: The integers are x, x + 2, x + 4, and x + 6. The condition is 18(x + 4) <= (x + x + 6) - 42. Thus 18x + 72 <= 2x - 36, so 16x <= -108 and x <= -6.75. Since x must be odd, the greatest possible value is -7.

Question 19
Skill: Scale Factors

A square map has side length 24 inches, and 1 inch on the map represents an actual distance of 15 miles. A smaller version of the same map is printed as a square with side length 40% shorter than the original. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 9

B) 15

C) 21

D) 25

Answer: D

Solution: The original side represents 24(15) = 360 miles. The smaller side length is 60% of 24, or 14.4 inches. The new scale is 360/14.4 = 25 miles per inch.

Question 20
Skill: Interpreting Exponential Function Forms

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$, and $b < 0$.

I. $f(x) = a(1.8)^{(x + b)}$
II. $g(x) = a(1.8)^x + b$

If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which equation displays, as a constant, the y-coordinate of the horizontal asymptote of the graph of the corresponding function?

A) I only

B) II only

C) I and II

D) Neither I nor II

Answer: B

Solution: For $f(x) = a(1.8)^{(x + b)}$, the horizontal asymptote is $y = 0$, which is not displayed as a constant in the equation. For $g(x) = a(1.8)^x + b$, the horizontal asymptote is $y = b$, and b is displayed as a constant. Therefore, II only.

Question 21
Skill: Nonlinear Functions and Zeros

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy-plane, the graph of $y = h(x)$ contains the points $(3, 0)$ and $(0, -\sqrt{360})$. If $h(m) = 0$, what is the greatest possible value of m?

Answer: 120

Solution: Since $h(3) = 0$, $x = 3$ is a zero of $x^2 + bx + c$. Since $h(0) = -\sqrt{360}$, $c = 360$. If the zeros are 3 and r, then their product is 360, so $3r = 360$ and $r = 120$. Therefore, the greatest possible value of m is 120.

Question 22
Skill: Absolute Value and Parameters

The function f is defined by $f(x) = |x|/a - 12$, where $a < 0$. What is the product of $f(11a)$ and $f(6a)$?

Answer: 414

Solution: Since $a < 0$, $|a| = -a$. Then $|11a|/a = -11$ and $|6a|/a = -6$. So $f(11a) = -11 - 12 = -23$ and $f(6a) = -6 - 12 = -18$. Their product is $(-23)(-18) = 414$.